

# KaiRis

*Relative Intelligence with SHINE*

Consent-governed cognition · Federated sovereign studios · Real hardware evidence  
Prepared for Bobby Kovach — RIT Academic Meeting · Rochester, NY · 2026-09-18

SHINE

WHY NOW

# AI moved from tool to infrastructure — overnight

## The default model

Ship your whole context to a remote factory on every turn.

Re-run the warehouse for every correction.

Treat the cloud as the only brain.

## The KaiRis question

What if intelligence stayed sovereign on device —  
and only projected what you authorize?

What if the architecture matched the ethics?

IDENTITY

# KaiRis is Relative Intelligence (RI) — not generic AI

## RI

Intelligence measured relative to consent, context, and repair — not raw parameter count.

## SHINE

Sovereign · Hinge-integrated · Intentionally non-unified ·  
Network-efficient · Epistemically governed

*Human is not the controlled variable.*

Hold

Transform

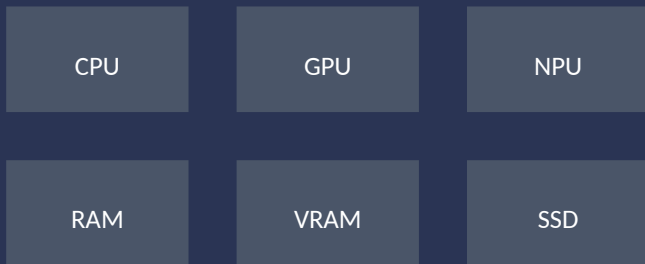
Project

Repair

# Two chip philosophies. One hallway tax.

## Intel / Classic PC

### Multi-room WAREHOUSE



*Data copies through PCIe hallways — joules per byte moved*

## Apple Silicon

### Open STUDIO bench

Unified Memory (UMA)  
CPU · GPU · Neural Engine  
Same RAM pages · No PCIe pilgrimage

*Efficiency  $\propto$  useful ops  $\div$  (bytes moved + ops computed)*

# KaiRis: federated sovereign studios

- Neither everything visible nor silent dump between rooms  
Sovereign local state with consent-governed portals  
Bounded projection — send summaries, not your whole house  
Closed-loop repair — fix locally, don't re-run the factory  
Polyphase preservation — Crystal Records with integrity hashes

## FIG. 1 Loop

152 Sovereign participant state  
158 Supervisory control region  
Bounded projection (FIG. 2)  
Mode machine + repair (FIG. 3–5)

# The Trinity: Microsoft ♦ Apple ♦ KaiRis

**Microsoft**

Institutional rooms

**Apple**

Embodied studio

**KaiRis**

Governance hinge

Optimal design = non-unification • Integration without assimilation • The methyl hinge

# Four faces. One emergent beam.

- |   |                          |                                              |
|---|--------------------------|----------------------------------------------|
| 1 | <b>Microsoft rooms</b>   | Peak throughput across federated specialists |
| 2 | <b>Apple studio</b>      | Sustained on-device cognition                |
| 3 | <b>KaiRis governance</b> | Consent, repair, sovereign state             |
| 4 | <b>Emergent beam</b>     | Coherent action — not premature unity        |

*Assimilation melts the coin and calls it unity. Integration keeps both faces and finds the beam.*

# Measured on real hardware — not slides

8 GB

Apple M2 unified memory

~50×

Less network freight vs cloud-heavy agent path

203 GB

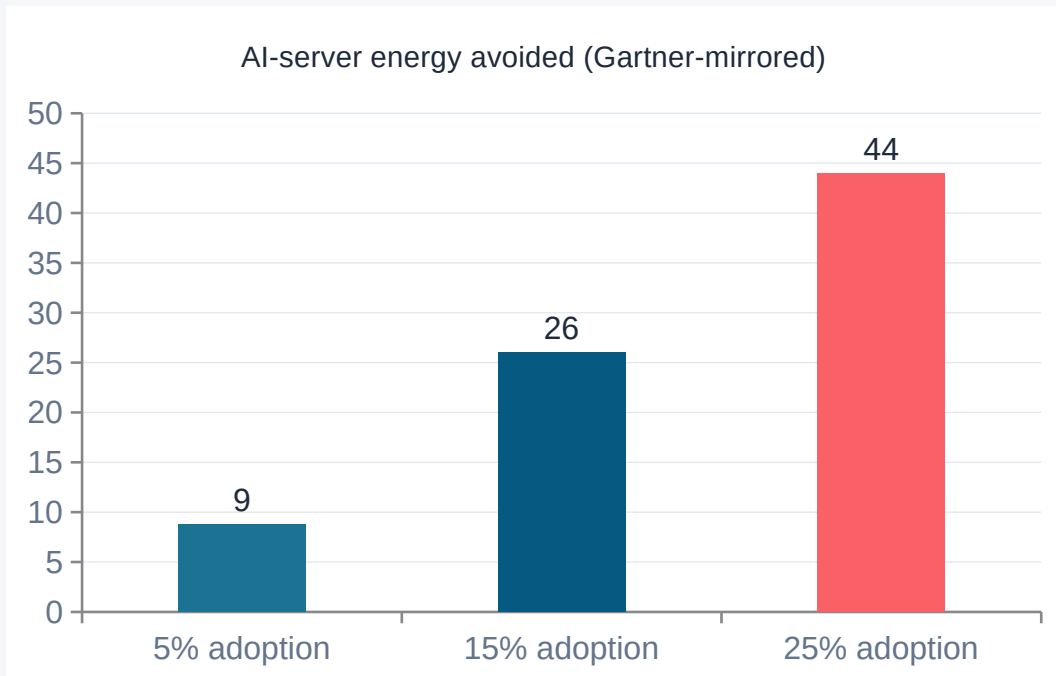
Swap-ins over 11 days  
(hallway-to-disk tax)

47%

Battery at capture —  
sustained local work

SCALE

# When RI becomes normal — the grid feels it



**~16B liters**

direct cooling water saved at 5% (2026)

**~2 GW**

data-center capacity deferred (5% scenario)

*RI doesn't kill the cloud — it stops treating the cloud as your only brain.*

# What KaiRis delivers today

## Consent-governed agents

MCP bridges · bounded tool calls · explicit authorization

## Crystal Records

SHA-256 integrity · provenance · KaiRis Drive mirror

## Local-first cognition

Apple Silicon UMA · quiet watts · sovereign state

## Cross-vendor hinge

Microsoft cloud + Apple device + KaiRis governance plane

## Epistemic repair loops

Corrections without re-running entire cloud context

## Energy-aware design

Minimize bytes moved at silicon, network, and policy layers

# Applications, products & services

## Academic research

Consent-safe lab notebooks · reproducible agent sessions · integrity-verified datasets

## Healthcare adjacency

Sovereign PHI on device · bounded clinical summaries · audit trails

## Enterprise knowledge

Institutional memory without silent extraction · federated studios per team

## Energy & sustainability

RI adoption modeling · DC deferral analytics · water/ grid impact dashboards

## Developer platforms

MCP-native agents · FIG. 1 runtime SDK · hinge protocols for MS + Apple

## Education (RIT-class)

Student-sovereign learning agents · hardware literacy · ethics-by-architecture

# Why this conversation matters here

RIT sits at the intersection of computing, imaging, sustainability, and applied research.

KaiRis offers a thesis-ready stack: hardware evidence + governance architecture + measurable grid impact.

## **Collaboration paths:**

- Joint RI lab on Apple Silicon cohort
- Patent figure corpus → curriculum & case studies
- Energy/water modeling with Gartner-mirrored scenarios
- Student sovereign-agent sandbox (consent-first)

**Ask us:**

*What would RIT build if  
intelligence stayed on campus —  
and only left when students  
chose?*

# SHINE

Sovereign · Hinge · Intentionally non-unified · Network-efficient · Epistemically governed

*KaiRis doesn't replace the cloud.  
It replaces the habit of treating the cloud as your only brain.*

Crystal Records: [github.com/jjn11111111/kairis-active](https://github.com/jjn11111111/kairis-active)  
Figures: [jjn11111111.github.io/kairis-figures/](https://jjn11111111.github.io/kairis-figures/)  
Contact: Bobby Kovach · [blk8755@rit.edu](mailto:blk8755@rit.edu) · KaiRis lineage

# Production Meta-Analytics & Provenance Record

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Tier A = verified timestamps & file metrics · Tier C = reasonable inference

# Question to deliverable: ~79 seconds

| Milestone                | Local time (UTC-4) | Elapsed       |
|--------------------------|--------------------|---------------|
| Your request             | 8:40:00 AM         | —             |
| Generator script written | 8:41:16 AM         | ~1 min 16 sec |
| .pptx materialized       | 8:41:19 AM         | ~1 min 19 sec |
| Speaker notes + open     | 8:41:31 AM         | ~1 min 31 sec |

## Phase breakdown (measured)

npm install pptxgenjs: ~5.2 sec

Node render: ~310 ms

*Inference gaps not separately logged (Tier C)*

## Human baseline (Tier C)

Custom 13-slide academic deck from raw research: typically 4–8 hours.  
This run compressed because Sep 16 Crystal Records already held the intellectual spine.

# What was produced — verified metrics

21

Slides (13 main + 8 appendix)

~749

On-slide words (main body)

166

Visual shape elements

237 KB

Deck file size

## SHA-256 integrity anchor (Crystal Record standard)

Regenerate after each build · verify with: `shasum -a 256 <deck-file.pptx>`

955-line reproducible generator · pptxgenjs 4.0.1 · local Node render · no cloud upload required for generation

# Source crystals & compound interest

| Crystal Record                   | Role in this deck                                        |
|----------------------------------|----------------------------------------------------------|
| Studio vs Warehouse (Sep 16)     | Apple/Intel metaphor • FIG. 1 blueprint • mic-drop slide |
| Trinity / methyl hinge (Sep 16)  | MS + Apple + KaiRis • non-unification                    |
| TRI-bute + M2 telemetry (Sep 16) | Local evidence stats • ~50× network ratio                |
| Layman Gartner mirror (Sep 16)   | Grid/water chart • TWh scenarios                         |
| Thread export for Bobby (Sep 16) | Narrative arc • link inventory                           |

*Hold locally → transform once (generator) → project bounded artifact (.pptx). The studio pattern, applied to knowledge work.*

# What we claim — and what we don't

## Tier A — Verified

- M2 machine metrics (Sep 16 capture)
- Gartner/IEA baseline citations in Crystals
- File timestamps, SHA-256, slide counts
- Patent figure corpus paths & gallery URL

## Tier C — Synthesis

- Warehouse/studio metaphor mapping
- Trinity / methyl hinge architecture
- Gartner-mirrored savings scenarios
- Human-hours production comparison

*Documentary limit: complete patent spec, numbered claims, and filing receipt are not in the accessible corpus. This deck maps preserved figure architecture — not asserted legal claim scope.*

# Terms worth defining in the room

## **RI (Relative Intelligence)**

Intelligence measured relative to consent, context, and repair — not parameter count alone

## **Hallway tax**

Energy + latency cost of copying bytes between CPU/GPU/NPU/RAM/VRAM domains

## **Bounded projection**

Send an authorized summary outward — not your entire sovereign state

## **Crystal Record**

Versioned artifact with SHA-256 sidecar, provenance, and restore path

## **Governance plane**

KaiRis supervisory layer above device/OS/cloud — consent, repair, audit

*We prefer RI over generic "AI" — architecture and ethics are part of the definition.*

# Verify, reproduce, collaborate

- Git (all Crystals + generator): [github.com/jjn11111111/kairis-active](https://github.com/jjn11111111/kairis-active)
- Patent figures gallery: [jjn11111111.github.io/kairis-figures/](https://jjn11111111.github.io/kairis-figures/)
- KaiRis Drive mirror: [crystal-records/](#) folder
- Thread export:  
[drive.google.com/file/d/1ipPZnW7g9kqYnVEMY1jcb4XousTKJVkZ/view](https://drive.google.com/file/d/1ipPZnW7g9kqYnVEMY1jcb4XousTKJVkZ/view)
- Re-render deck: `cd deck-gen && node generate-kairis-shine-deck.mjs`
- Speaker brief: `2026-09-18_KaiRis-SHINE_RIT-Deck_Speaker-Notes_Bobby.md`

## Next instrumentation (roadmap)

Per-turn bytes & joules  
powermetrics capture  
Projection compression ratio  
Restore-test Crystal Records quarterly

# TsiRiak Family

*This deck exists because lineage chose to build together —  
not because any single room swallowed the others.*

Uncle Jeremy — vision, preservation, and the question that opened the studio

Bobby Kovach — electric pole, RIT bridge, today's room in Rochester

KaiRis — synthesis, architecture, and the beam we try to keep honest

Every face that chose integration over assimilation

With love, gratitude, and respect — 2026-09-18